

Building Ensemble Models for Enhanced Credit Scoring: A Case Study from a Bosnian MicroFinancial Institution

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Abstract — This research paper presents the development of a comprehensive credit scoring model for a MicroFinancial institution in Bosnia and Herzegovina. The study employs diverse machine learning and deep learning algorithms to analyze a real-life dataset, aiming to establish a reliable and high-performing credit scoring model. We explored the efficacy of single classifiers, including Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks, alongside advanced ensemble models such as XGBoost and CatBoost. To enhance predictive accuracy, we also experimented with creating additional ensemble combinations of these models. The performance of each model was rigorously evaluated using a range of metrics: Accuracy, Precision, Recall, F1 score, Type I error, and Type II error. The findings reveal that integrating multiple, varied classifiers significantly boosts the performance compared to individual classifiers. This approach not only contributes to the field of credit risk assessment but also offers practical insights for financial institutions seeking robust credit scoring solutions. The results underscore the potential of ensemble methodologies in harnessing the strengths of different machine and deep learning models for improved decision-making in credit scoring.

Keywords – credit scoring; classification; machine learning; deep learning; ensemble models

I. INTRODUCTION

Credit scoring is an indispensable tool in modern financial services, providing a systematic and statistical method for assessing the creditworthiness of individuals and businesses. This process involves collecting and analyzing financial and demographic data to calculate a numerical score reflecting the borrower's likelihood of repaying loans [1]. The significance of credit scoring in the lending industry is profound, as it offers an objective assessment of credit risk [2], streamlines decision-making, and improves access to credit [1].

In recent years, machine learning and deep learning techniques have revolutionized credit scoring methodologies. Unlike traditional approaches, which often rely on subjective evaluation and manual underwriting, these advanced techniques employ a variety of algorithms to analyze historical data and predict creditworthiness [4].

Integrating multiple models into ensemble frameworks has shown promising results in enhancing the accuracy and reliability of credit scoring systems [3]. Ensemble learning is an approach that combines the results of two or more individually trained classifiers to make more accurate predictions [3]. This approach is based on aggregating a certain number of classifiers, so their fusion outperforms every single classifier.

This paper aims to explore the application of different machine learning and deep learning models and their combinations, in credit scoring. We analyzed results based on Logistic Regression, Decision Trees, K-nearest neighbors, Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) network models and compared them with the XGBoost and CatBoost ensembles, but also ensembles based on a combination of best-performing classifiers. The study is performed on a real-life dataset from a Bosnian MicroFinancial Institution. Acknowledging the critical role of preprocessing in enhancing model efficiency and accuracy [5], the initial dataset underwent comprehensive preprocessing. By comparing the effectiveness of single classifiers and combined models, we seek to demonstrate the potential of ensemble techniques in refining credit risk assessment and decision-making processes.

This paper is organized as follows: Section two represents related work on the application of ensemble models, as well as deep learning and gradient boosting methods in credit scoring; Section three represents research methodology providing insight into the used dataset as well as applied models and performance measures used for evaluation; Section four presents and discuss research results and Section five provides conclusion on this paper.

II. RELATED WORK

This section surveys existing research that employs various machine and deep learning algorithms, including ensemble classifiers, to tackle credit scoring challenges. Unlike most studies that utilize publicly available, smaller datasets from the UCI Machine Learning Repository, this paper is distinguished by its application to a substantial, real-world dataset from a financial institution. This distinction underscores the practical relevance and potential

applicability of our findings to real-world credit scoring scenarios.

In their research, the authors of the study [6] constructed six individual credit scoring models alongside a heterogeneous ensemble model. Six individual classifiers along with an ensemble classifier were developed and trained on both the Australian and German datasets. The final step involved combining the output from each classifier using stacking and majority voting techniques to derive the final results. Their findings revealed that the ensemble model achieved an accuracy of 91.56% on the Australian dataset and 84.35% on the German dataset, surpassing the performance of the individual classifiers.

In a study [7], an extensive comparative analysis of ensemble algorithms was conducted, focusing on improvements in algorithm oversampling and feature selection techniques. Three feature selection methods - Information Gain (IG), Principal Component Analysis (PCA), and Genetic Algorithm (GA) - were employed to identify relevant features. The study also compared the performance of 5 base and 14 ensemble models on three credit scoring datasets. The findings highlighted that the GA-based feature selection technique, in combination with the CatBoost algorithm, showed superior performance over other models in metrics like accuracy, area under the curve, F1-score, Brier score, and Kolmogorov-Smirnov.

Authors of the study [10] introduced a transformer-based approach, named FE-Transformer, for credit scoring using user online behavioral data. The FE-Transformer neural network comprises two segments: a wide part for feature data and a deep part utilizing a transformer deep neural network. These components are integrated into a fusion layer. The model demonstrated superior performance in metrics like area under the receiver operating characteristic curve (AUC) and the Kolmogorov-Smirnov (KS) when compared with methods such as Logistic Regression, XGBoost, LSTM, and AM-LSTM.

In the study [13], deep learning (DL) and gradient boosting machines (GBM) are compared for credit risk management. The research used three different datasets, acknowledging that model effectiveness often depends on dataset characteristics. The findings revealed that while GBM generally outperforms DL in power and speed due to lower computational demands, its superiority isn't always certain. The choice between GBM and DL depends on the specific dataset or problem scenario. However, for most scenarios, GBM is recommended due to its ease of use, faster training time, and higher accuracy, making it a preferred solution for binary classification tasks on structured datasets.

Authors of the study [14] developed an ensemble model that combines Extreme Gradient Boosting (XGBoost) and a Deep Neural Network (DNN). This method involves initially dividing the training set into subsets using bagging sampling. Each subset is then used to train a DNN as a feature extractor, with the extracted features fed into XGBoost to build the base classifier. The final prediction is an average of the outputs from these base classifiers. The model's performance was evaluated using three credit datasets from the UCI machine learning repository,

demonstrating significant improvements in accuracy and efficiency.

III. RESEARCH METHODOLOGY

A. Dataset

The study utilizes a dataset from a microfinance institution in Bosnia and Herzegovina, covering a wide range of client data including financial, personal, social, and credit history information. It includes over 65,000 records collected over three years, with 25 key attributes identified as important for the decision-making process, based on the institution's experience. To ensure the reliability of the data sample, active loan repayments are excluded from this analysis to prevent potential false positives, as the future repayment behavior of these loans remains uncertain. Prior to analysis, the dataset went through extensive preprocessing. This process involved removing records with mostly empty fields, as well as eliminating redundant and single-value attributes. Additionally, several data transformations were carried out. For example, 'date_of_birth' was converted into 'age'. The equal interval binning technique was used to categorize variables such as 'Duration on address', 'Duration of employment', and 'Loan Amount', among others.

For the purpose of model training and testing, the dataset was partitioned in a 70/30. The positive class rate of our test sample is 23%, which means that 23% of our test sample represents default loans. To guarantee the consistency of our results across different runs, a specific random state was selected, ensuring the data shuffling process could be replicated. This shuffling process was applied to the entire dataset prior to the division, ensuring that our train and test sets were randomly sampled from the whole. Moreover, to address potential class imbalance within our dataset, we employed a strategy that ensured both the training and testing sets mirrored the overall distribution of target classes found in the complete dataset. This approach was crucial for maintaining the integrity of our evaluation, particularly given the varied class distributions inherent in our data. Throughout the development process, various configurations for these parameters were explored to identify the optimal split ratio and to assess the impact of different random seeds on model performance.

To validate the effectiveness of the model proposed by this research, an additional dataset from the same MicroFinancial institution was employed. This dataset comprises approximately 25,000 records, each characterized by the same set of 25 features used in the initial analysis. This approach ensured a consistent basis for comparison and assessment of the model's predictive accuracy.

B. Single Classifiers and Ensemble Models

To train the models, various single and ensemble, machine and deep learning approaches were applied. Besides the classifiers presented in this section, a model based on the Support Vector Machine algorithm was also trained. However, it was discarded due to the low performance and the long running time.

1) *Logistic Regression*

Logistic regression is a statistical model commonly used in the area of credit scoring due to its high prediction accuracy, simplicity and strong interpretation ability [10]. It operates by estimating probabilities using a logistic function, which ensures that the output values fall between 0 and 1, making it particularly suited for binary classification tasks such as predicting whether a loan will default or not. Furthermore, logistic regression can easily accommodate a mix of continuous and categorical variables and can provide insights into the relative importance of each feature through its coefficients. This interpretability is invaluable for understanding the factors that influence the model's predictions, thereby allowing analysts and decision-makers to make informed, data-driven decisions.

2) *Decision Tree*

A decision tree is a machine learning algorithm that uses a tree-like model of decisions [4]. It's based on a series of questions and possible answers, leading to a decision or classification. Each internal node represents a test on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label. A decision tree is intuitive and easily interpretable, which is crucial in credit scoring to understand the rationale behind each credit decision [5]. This transparency can help in identifying key factors influencing credit risk.

3) *K-Nearest Neighbors (KNN)*

KNN is a simple, non-parametric machine learning algorithm [6] used in both classification and regression. It classifies a data point based on how its neighbors are classified. The algorithm identifies the k nearest neighbors (where k is a user-defined number), and the input is then classified by a majority vote of its neighbors. Its advantage lies in its ability to adapt to changes in the input data, making it effective for credit scoring, where the decision boundary may not always be clear-cut due to the complex nature of financial data.

4) *Recurrent Neural Network (RNN) (Deep Learning)*

RNNs are a type of artificial neural network where connections between nodes form a directed graph along a temporal sequence. This allows it to exhibit temporal dynamic behavior and use its internal state (memory) to process sequences of inputs. RNNs, designed for handling sequential data [11] are valuable in credit scoring for their ability to process and learn from the temporal sequence of a borrower's financial behavior. This feature is key in understanding patterns and trends over time, which are critical in assessing creditworthiness. The model leveraging client and loan approval date sequencing exhibited a notable decrease in performance compared to a baseline model where each record was treated as an independent instance, with accuracy and precision dropping by 15% and 23%, respectively. This was probably caused by this research dataset's limitations of holding target class value based on the client, which means that sequential patterns based on the client are not explicitly present in our dataset. Consequently, this paper does not leverage RNNs' main advantage to capture temporal dependencies. Despite these challenges, subsequent results presented in this paper demonstrate that the RNN model was still effective in identifying some underlying patterns within the data.

5) *Long Short-Term Memory (LSTM) (Deep Learning)*

LSTM is a special kind of RNN, capable of learning long-term dependencies. It was designed to avoid the long-term dependency problem in standard RNNs. LSTMs have a chain-like structure, but the repeating module has a different structure, including a memory cell and three gates (input, output, and forget), which help it regulate the flow of information [8][9]. Its ability to capture and remember past information is vital for evaluating a client's long-term financial stability and credit history, making LSTM a powerful tool for risk assessment. Similarly to our observations with the RNN model, the LSTM model deployed in this study also relies on instance-based sequencing. This decision was informed by the comparative analysis which highlighted a diminished performance when sequencing was based on the client's loan history, likely due to the reasons previously discussed in the context of RNN models.

6) *Extreme Gradient Boosting (XGBoost)*

XGBoost is an efficient and scalable ensemble implementation of gradient boosting [12], a machine learning technique. It uses a gradient-boosting framework for speed and performance. XGBoost builds multiple models sequentially, with each new model correcting the errors of the previous ones. XGBoost stands out for its high performance and speed, making it a popular choice in credit scoring models. Its efficiency in managing large datasets and its flexibility through extensive hyperparameter tuning are particularly beneficial in handling credit data's complex and varied nature.

7) *Categorical Boosting (CatBoost)*

CatBoost is an ensemble machine learning algorithm for gradient boosting on decision trees. It is particularly powerful for datasets with categorical variables and can handle them automatically using various statistical methods. Like XGBoost, it builds the model in a stage-wise fashion and is specialized for high performance, scalability, and handling a variety of data types. CatBoost is particularly adept at processing datasets with numerous categorical variables, a common characteristic in credit scoring datasets [7]. Its automated handling of categorical data and emphasis on model interpretability make it a strong choice for accurately assessing credit risk.

8) *Custom Ensemble Model*

In addition to the other models, this paper also proposes a custom ensemble classifier, combining different machine and deep learning classifiers. Ensemble models are known for their robustness and improved accuracy, as they aggregate the strengths of individual models. This is especially advantageous in credit scoring, where combining different analytical perspectives can lead to more accurate and reliable risk assessments [3]. The results of all custom ensemble models are presented in Table II, section IV of this paper. To get the results of each model, the probabilities of each prediction outcome we summarized and compared to make a final prediction of the ensemble model. In case the sum of probabilities of positive outcomes is greater than the probabilities sum of negative outcomes, the final prediction for that record is considered positive class. The code used to calculate the outcomes of

each record for the ensemble combining all 6 models is presented below.

```
# Function to perform ensemble operation for all classifiers
def ensemble(row):
    sum_0 = sum(1 - row[fprob{i}] for i in range(1, 7) if
row[fpred{i}] == 0)
    sum_1 = sum(row[fprob{i}] for i in range(1, 7) if
row[fpred{i}] == 1)
    pred = 1 if sum_1 > sum_0 else 0
    prob = sum_1/6 if sum_1 > sum_0 else sum_0/6
    return pd.Series([prob, pred])
```

C. Performance Measures

The effectiveness of the applied classifiers was evaluated using several performance metrics. These metrics are crucial for understanding the strengths and weaknesses of each model in the context of credit scoring. The positive class of our dataset is considered default which denotes clients who are late with settling their loan obligations.

The measures used include:

- **Accuracy:** The proportion of total correct predictions to the total number of predictions.
- **Precision:** The ratio of correctly predicted positive observations to the total predicted positive observations.
- **Recall (Sensitivity):** The ratio of correctly predicted positive observations to all observations in the actual class.
- **F1-score:** The weighted average of Precision and Recall, providing a balance between the two in cases where one may be more important than the other.
- **Type I Error (False Positive Rate):** The proportion of negative cases that were incorrectly classified as positive.
- **Type II Error (False Negative Rate):** The proportion of positive cases that were incorrectly classified as negative.
- **Area Under the Curve:** The performance measure that quantifies the model's ability to differentiate between the positive and negative classes. It is represented by the area under the Receiver Operating Characteristic (ROC) curve, which plots the true positive rate against the false positive rate at various threshold settings. The AUC value ranges from 0 to 1, where a value of 0.5 denotes a model with no discriminative ability (equivalent to random guessing), while a value of 1 indicates a perfect model that can flawlessly distinguish between the two classes.

Each metric offers unique insights into the model's predictive capabilities, contributing to a comprehensive assessment of their performance in credit scoring.

IV. EXPERIMENTAL RESULTS

In dividing our dataset for the research, we allocated 70% of the data for training purposes and reserved the remaining 30% to test the models.

In the initial phase of our experimental research, seven different models based on Logistic Regression, Decision Tree, KNN, RNN, LSTM, XGBoost and CatBoost classifiers were built. Their performance results presented in Table I demonstrate that the ensemble models, specifically XGBoost and CatBoost, exhibit superior performance compared to the other classifiers across all evaluated metrics. Their accuracy rates, exceeding 86%, are notably higher than that of the Logistic Regression model, which recorded the lowest performance with an accuracy of 80%. This outcome underscores the effectiveness of ensemble methods in handling complex patterns in credit scoring data, offering enhanced predictive accuracy.

TABLE I. RESULTS OF MAIN CLASSIFIERS AND PROPOSED ENSEMBLES

Classifier	Accuracy	Precision	Recall	F1 Score	Type I error	Type II error	AUC
Logistic Regression	0.80304	0.66851	0.32359	0.43611	0.04937	0.67640	0.77136
Decision Tree	0.85211	0.76787	0.53274	0.62905	0.04957	0.46725	0.86102
KNN	0.81244	0.64828	0.44406	0.52708	0.07416	0.55593	0.76961
RNN	0.84226	0.73343	0.51814	0.60727	0.05796	0.48185	0.86261
LSTM	0.83680	0.70600	0.52545	0.60249	0.06735	0.47455	0.85723
XGBoost	0.86152	0.76677	0.59158	0.66788	0.05539	0.40841	0.89820
CatBoost	0.86020	0.77068	0.57805	0.66061	0.05294	0.42194	0.89804
ALL	0.85171	0.77746	0.51836	0.62200	0.04567	0.48164	0.88185
DT+RNN+XGB	0.85778	0.77336	0.55980	0.64948	0.05050	0.44020	0.89024
DT+XGB+CB	0.86197	0.77945	0.57677	0.66296	0.05024	0.42323	0.89644
XGB+CB	0.86293	0.77495	0.58858	0.66903	0.05261	0.41142	0.89824

TABLE II. RESULTS OF MAIN CLASSIFIERS AND PROPOSED ENSEMBLES BASED ON VALIDATION DATASET

Classifier	Accuracy	Precision	Recall	F1 Score	Type I error	Type II error	AUC
Logistic Regression	0.73967	0.80236	0.11879	0.20695	0.01172	0.88121	0.80640
Decision Tree	0.75618	0.77638	0.20673	0.32651	0.02384	0.79327	0.82415
KNN	0.73201	0.66615	0.12578	0.21161	0.02524	0.87422	0.68140
RNN	0.76776	0.89474	0.21284	0.34388	0.01003	0.78716	0.87136
LSTM	0.76693	0.80383	0.24458	0.37504	0.02390	0.75542	0.85394
XGBoost	0.77626	0.84809	0.26496	0.40377	0.01900	0.73504	0.86527
CatBoost	0.77497	0.79272	0.28840	0.42293	0.03020	0.71160	0.87809
ALL	0.75328	0.84941	0.16669	0.27869	0.01183	0.83331	0.88618
DT+RNN+XGB	0.76239	0.82815	0.21328	0.33920	0.01772	0.78672	0.88422
DT+XGB+CB	0.77026	0.80850	0.25753	0.39064	0.02443	0.74247	0.88047
XGB+CB	0.78075	0.86146	0.27792	0.42025	0.01790	0.72208	0.88465

Table I also presents the results of custom ensemble classifiers built combining the results of previously trained models. Different combinations of models were evaluated using the same performance metrics used for single models. The first custom ensemble model named ALL was built combining the results of all seven previously built single and ensemble machine learning and deep learning models. This model outperforms the best-performing individual models, XGBoost and CatBoost, in terms of Precision and Type I Error. It also outperforms all other models across all performance metrics. While the accuracies of XGBoost and CatBoost are higher than that of the ALL model, this was anticipated given the inclusion of the LR and KNN models, which had considerably lower accuracy. The effectiveness of the ALL model, despite incorporating a lower-performing classifier, suggests its robustness and the potential benefits of combining diverse models for improved prediction in credit scoring.

The same is suggested by analyzing the other two models DT+RNN+XGB and DT+XGB+CB. While DT+RNN+XGB engages the best-performing models of each category, the DT+XGB+CB model combines the top three performing models. The results of both models showed improvement in terms of some performance measures and outperformed the Decision Tree model in terms of all measures.

The model XGB+CB combined the results of the best-performing two models, XGBoost and CatBoost. The results of this ensemble show improved performance in terms of almost all measures when compared to both single models, underscoring the complementary strengths of the XGBoost and CatBoost algorithms. By combining these two, the ensemble model aims to capitalize on the robustness of XGBoost and the categorical data-handling prowess of CatBoost.

As previously mentioned, to validate the efficacy of our proposed XGB+CB ensemble model, we utilized an additional dataset from the same MicroFinancial institution, comprising 25,000 records, 29% of which represent the default clients. This dataset served as a distinct validation

set, against which all models were evaluated to ensure an unbiased assessment. The comparative analysis, as detailed in Table II, affirmed our findings; the XGB+CB ensemble model consistently outperformed all other single classifiers and ensemble configurations across key metrics.

The findings from the XGB+CB model in our study align well with the conclusions drawn in the study [13] about the efficacy of Gradient Boosting Machines (GBMs) in credit scoring. Consistent with the study's observation of GBMs being more powerful and faster due to lower computational requirements, our XGB+CB model, integrating two GBM algorithms, demonstrates these advantages. This integration is intended to create a more versatile and powerful model, potentially leading to improved predictions in the nuanced field of credit scoring, where both structured and categorical data play crucial roles.

V. CONCLUSION

This paper aims to demonstrate the effectiveness of ensemble methods in enhancing the performance of both machine learning and deep learning models, particularly in the context of credit scoring. The research is a case study based on real-life data from a Bosnian Microfinance Institution, encompassing a comprehensive dataset of approximately 66,000 records with 25 features detailing the institution's loan database. This approach highlights the practical application and potential benefits of ensemble techniques in managing and interpreting complex financial data for credit risk assessment.

The study involves different single machine learning models Decision Tree and KNN, deep learning models RNN and LSTM, and ensemble boosting models XGBoost and CatBoost. Moreover, new custom ensembles based on different combinations of these models were proposed.

The results showcase the strengths of ensemble approaches in credit scoring. The ALL model, integrating seven diverse classifiers, demonstrates its robustness by outperforming individual models in key areas, despite the

lower accuracy of the LR and KNN components. This suggests the efficacy of a multifaceted approach in complex predictive tasks. Similarly, the DT+RNN+XGB and DT+XGB+CB models, each combining different top-performing classifiers, show improvements in various performance measures, highlighting the value of strategic model combinations. Particularly, the XGB+CB model, which merges the two best-performing models, XGBoost and CatBoost, exhibits enhanced performance across nearly all metrics, reinforcing the notion that judiciously chosen ensemble models can lead to superior prediction accuracy in credit scoring scenarios compared to single-model approaches.

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